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PHOTOELECTRIC CONVERSION DEVICE, INFORMATION PROCESSING DEVICE, METHOD FOR CONTROLLING PHOTOELECTRIC CONVERSION DEVICE, AND INFORMATION PROCESSING METHOD

Abstract

A photoelectric conversion device includes a pixel, a memory, and a processing unit. The pixel generates the signal in a second accumulation period including a first accumulation period and a subsequent period. The processing unit acquires a first signal corresponding to an amount of incident light in the first accumulation period from the memory in a period from an end of the first accumulation period to an end of the second accumulation period. The processing unit generates pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using the first signal. The processing unit acquires a second signal corresponding to an amount of incident light in the second accumulation period from the memory after the end of the second accumulation period. The processing unit corrects the second signal based on the pixel information.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to a photoelectric conversion device, an information processing device, a method for controlling a photoelectric conversion device, and an information processing method.

Description of the Related Art

[0002] Japanese Patent Application Laid-Open No. 2008-148129 discloses an imaging device that detects an abnormal pixel in each of a plurality of images captured under different conditions, and determines an abnormality factor of the detected abnormal pixel based on a difference in output levels of the abnormal pixels.

[0003] However, there is a case where it is required to further reduce a processing delay caused by detecting and correcting a pixel that is outside a predetermined reference range.

SUMMARY OF THE INVENTION

[0004] An object of the present disclosure is to provide a photoelectric conversion device, an information processing device, a method for controlling a photoelectric conversion device, and an information processing method in which a delay time for detecting and correcting a pixel that is outside a predetermined reference range is further reduced.

[0005] According to one disclosure of the present specification, there is provided a photoelectric conversion device including a pixel configured to generate a signal corresponding to an amount of incident light, a memory configured to hold the signal, and a processing unit configured to perform signal processing based on the signal held in the memory. The pixel generates the signal in a second accumulation period including a first accumulation period and a period after the first accumulation period. The processing unit acquires a first signal corresponding to an amount of incident light in the first accumulation period from the memory in a period from an end of the first accumulation period to an end of the second accumulation period. The processing unit generates pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using the first signal. The processing unit acquires a second signal corresponding to an amount of incident light in the second accumulation period from the memory after the end of the second accumulation period. The processing unit corrects the second signal based on the pixel information.

[0006] According to one disclosure of the present specification, there is provided an information processing device including a detection unit configured to generate pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using a first signal corresponding to an amount of light incident on the pixel in a first accumulation period, and a correction unit configured to correct a second signal corresponding to an amount of light incident on the pixel in a second accumulation period including the first accumulation period and a period after the first accumulation period based on the pixel information. The detection unit acquires the first signal during a period from an end of the first accumulation period to an end of the second accumulation period. The correction unit acquires the second signal after the end of the second accumulation period.

[0007] According to one disclosure of the present specification, there is provided a method for controlling a photoelectric conversion device including a pixel configured to generate a signal corresponding to an amount of incident light, a memory configured to hold the signal, and a processing unit configured to perform signal processing based on the signal held in the memory. The method includes generating, by the pixel, the signal in a second accumulation period including a first accumulation period and a period after the first accumulation period, acquiring, by the processing unit, a first signal corresponding to an amount of incident light in the first accumulation period from the memory in a period from an end of the first accumulation period to an end of the second accumulation period, generating, by the processing unit, pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using the first signal, acquiring, by the processing unit, a second signal corresponding to an amount of incident light in the second accumulation period from the memory after the end of the second accumulation period, and correcting, by the processing unit, the second signal based on the pixel information.

[0008] According to one disclosure of the present specification, there is provided an information processing method including generating abnormal pixel information indicating an abnormal pixel detection result based on a first signal corresponding to an amount of light incident on a pixel in a first accumulation period, and correcting a second signal corresponding to an amount of light incident on the pixel in a second accumulation period including the first accumulation period and a period after the first accumulation period based on the abnormal pixel information. The first signal is acquired during a period from an end of the first accumulation period to an end of the second accumulation period. The second signal is acquired after the end of the second accumulation period.

[0009] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a schematic diagram illustrating an overall configuration of a photoelectric conversion device according to a first embodiment.

[0011] FIG. **2** is a schematic block diagram illustrating a configuration example of a sensor substrate according to the first embodiment.

[0012] FIG. **3** is a schematic block diagram illustrating a configuration example of a circuit substrate according to the first embodiment.

[0013] FIG. **4** is a schematic block diagram illustrating a configuration example for one pixel of a photoelectric conversion unit and a pixel signal processing unit according to the first embodiment.

[0014] FIGS. **5**A, **5**B, and **5**C are diagrams for explaining an operation of an avalanche photodiode according to the first embodiment.

[0015] FIG. **6** is a block diagram illustrating a configuration example of a pulse processing unit according to the first embodiment.

[0016] FIG. **7** is a timing chart for explaining an operation of the pulse processing unit according to the first embodiment.

[0017] FIG. **8** is a diagram illustrating an example of an image acquired by the photoelectric conversion device according to the first embodiment.

[0018] FIG. **9** is a block diagram illustrating a configuration example of a signal processing unit according to the first embodiment.

[0019] FIG. **10** is a schematic diagram illustrating an example of a target pixel and peripheral pixels according to the first embodiment.

[0020] FIG. **11** is a timing chart for explaining an operation of the signal processing unit according to the first embodiment.

[0021] FIG. **12** is a schematic diagram illustrating an example of a target pixel and peripheral pixels according to a modification of the first embodiment.
[0022] FIG. **13** is a block diagram illustrating a configuration example of a signal processing unit according to a second embodiment.
[0023] FIG. **14** is a timing chart for explaining an operation of the signal processing unit according to the second embodiment.
[0024] FIG. **15** is a block diagram illustrating a configuration example of a signal processing unit according to a third embodiment.
[0025] FIG. **16** is a timing chart for explaining an operation of the signal processing unit according to the third embodiment.
[0026] FIG. **17** is a block diagram illustrating a configuration example of a signal processing unit according to a fourth embodiment.
[0027] FIG. **18** is a flowchart for explaining an operation of a photoelectric conversion device according to the fourth embodiment.
[0028] FIG. **19** is a block diagram of equipment according to a fifth embodiment.
[0029] FIGS. **20**A and **20**B are block diagrams of equipment according to a sixth embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0030] Preferred embodiments of the present invention will now be described in detail in accordance with the accompanying drawings. Identical elements or corresponding elements across multiple drawings are denoted by common reference signs, and the description thereof may be omitted or simplified.

First Embodiment

[0031] FIG. **1** is a schematic diagram illustrating an overall configuration of a photoelectric conversion device according to the present embodiment. The photoelectric conversion device **100** includes a sensor substrate **11** and a circuit substrate **21** stacked on each other. The sensor substrate **11** and the circuit substrate **21** are electrically connected to each other. The sensor substrate **11** has a pixel region **12** in which a plurality of pixels **101** are arranged to form a plurality of rows and a plurality of columns. The circuit substrate **21** includes a first circuit region **22** in which a plurality of pixel signal processing units **103** are arranged to form a plurality of rows and a plurality of columns, and a second circuit region **23** arranged on an outer periphery of the first circuit region **22**. The second circuit region **23** may include a circuit or the like that controls the plurality of pixel signal processing units **103**.
[0032] FIG. **2** is a schematic block diagram illustrating a configuration example of the sensor substrate **11** according to the present embodiment. In the pixel region **12**, a plurality of pixels **101** are arranged to form a plurality of rows and a plurality of columns. Each of the plurality of pixels **101** includes a photoelectric conversion unit **102** including an avalanche photodiode (hereinafter, referred to as APD) as a photoelectric conversion element in the substrate. FIG. **2** illustrates some of m×n pixels **101** arranged in m rows from the first row to the m-th row and n columns from the first column to the n-th column, together with reference signs indicating row numbers and column numbers. For example, the pixel **101** arranged in the first row and the third column is denoted by reference sign “P13”. Note that the number of rows and the number of columns of the pixel array constituting the pixel region **12** are not particularly limited.
[0033] FIG. **3** is a schematic block diagram illustrating a configuration example of the circuit substrate **21** according to the present embodiment. The circuit substrate **21** includes a first circuit region **22**, a second circuit region **23**, a signal reading circuit **115**, and a signal processing unit **117** (processing unit).
[0034] In the first circuit region **22**, the plurality of pixel signal processing units **103** are arranged to form a plurality of rows and a plurality of columns. In FIG. **3**, some of the m×n pixel signal processing units **103** arranged in m rows from the first row to the m-th row and n columns from the first column to the n-th column, together with reference signs indicating row numbers and column

numbers. For example, the pixel signal processing unit **103** arranged in the first row and the third column is denoted by reference signs “S13”. Note that the number of rows and the number of columns of the pixel signal processing unit array constituting the first circuit region **22** are not particularly limited.

[0035] In the second circuit region **23**, a vertical control pulse generation unit **110** and a horizontal control pulse generation unit **111** are arranged. In each row of the pixel signal processing unit array in the first circuit region **22**, a vertical control line **112** and a data output line **114** extending in a first direction (a transverse direction in FIG. **3**) are arranged. The vertical control line **112** and the data output line **114** are connected to each of the plurality of pixel signal processing units **103** arranged in the first direction. The first direction in which the vertical control line **112** extends may be referred to as a row direction or a horizontal direction.

[0036] In each column of the pixel signal processing unit array in the first circuit region **22**, a horizontal control line **113** extending in a second direction (a longitudinal direction in FIG. **3**) is arranged. The horizontal control line **113** is connected to each of the plurality of pixel signal processing units **103** arranged in the second direction. The second direction in which the horizontal control line **113** extends may be referred to as a column direction or a vertical direction.

[0037] The vertical control line **112** for each row is connected to the vertical control pulse generation unit **110**. The vertical control pulse generation unit **110** supplies a control signal for driving the pixel signal processing unit **103** to the pixel signal processing unit **103** via the vertical control line **112**. The horizontal control line **113** for each column is connected to the horizontal control pulse generation unit **111**. The horizontal control pulse generation unit **111** supplies a control signal for driving the pixel signal processing unit **103** to the pixel signal processing unit **103** via the horizontal control line **113**. The vertical control pulse generation unit **110**, the horizontal control pulse generation unit **111**, and the signal reading circuit **115** are connected via a read control line **116**. The vertical control pulse generation unit **110** supplies a control signal for driving the horizontal control pulse generation unit **111** and the signal reading circuit **115** via the read control line **116**. As a result, the vertical control pulse generation unit **110**, the horizontal control pulse generation unit **111**, and the signal reading circuit **115** operate in a synchronized state. The vertical control pulse generation unit **110** may generate a control signal based on an external trigger that is not illustrated, or may generate a control signal based on an internal signal.

[0038] The data output line **114** for each row is connected to the signal reading circuit **115**. The data output line **114** is a signal line for transmitting data held by the pixel signal processing unit **103**. In response to the control signal supplied from the vertical control pulse generation unit **110** via the read control line **116**, the signal reading circuit **115** acquires a plurality of pieces of data from the data output line **114** and outputs the acquired data to the signal processing unit **117**.

[0039] The signal processing unit **117** is an information processing device that processes the acquired image data. The signal processing unit **117** reads and executes a computer-executable instruction. The signal processing unit **117** may be a computer including one or more processors and one or more memories. The signal processing unit **117** may include a plurality of separated computers, or may include a plurality of separated processors. Furthermore, the signal processing unit **117** may include one or more processing circuits. The processor or circuit may include a central processing unit (CPU), a micro processing unit (MPU), a graphics processing unit (GPU), an application specific integrated circuit (ASIC), or a field programmable gate array (FPGA). The processor or circuit may also include a digital signal processor (DSP), a data flow processor (DFP), or a neural processing unit (NPU).

[0040] Although the signal processing unit **117** is arranged in the circuit substrate **21** in the example of the present embodiment, the function of the signal processing unit **117** may be arranged outside the circuit substrate **21**. For example, the function of the signal processing unit **117** may be arranged on an external controller system. In this case, data is output from the signal reading circuit **115** to the controller system via an output interface circuit. That is, in the example of the present embodiment, the photoelectric conversion device **100** is an integrated device including the signal processing unit **117**, but may be configured as a photoelectric conversion system including a photoelectric conversion device and a controller system (information processing device) communicatively connected to each other.

[0041] FIG. **4** is a schematic block diagram illustrating a configuration example for one pixel of the photoelectric conversion unit **102** and the pixel signal processing unit **103** according to the present embodiment. FIG. **4** schematically illustrates a more specific configuration example including a connection relationship between the photoelectric conversion unit **102** arranged in the sensor substrate **11** and the pixel signal processing unit **103** arranged in the circuit substrate **21**.

[0042] The photoelectric conversion unit **102** includes an APD **201**. The pixel signal processing unit **103** includes a pulse generation unit **210** and a pulse processing unit **220**. The pulse generation unit **210** includes a quenching element **211** and a waveform shaping unit **212**.

[0043] The APD **201** generates a charge pair corresponding to incident light by photoelectric conversion. A voltage VL (first voltage) is supplied to an anode of the APD **201**. In addition, a cathode of the APD **201** is connected to a first terminal of the quenching element **211** and an input terminal of the waveform shaping unit **212**. A voltage VH (second voltage) higher than the voltage VL supplied to the anode is supplied to the cathode of the APD **201**. As a result, a reverse bias voltage is supplied to the anode and the cathode of the APD **201** such that the APD **201** performs an avalanche multiplication operation. In the APD **201** to which the reverse bias voltage is supplied, when a charge is generated by incident light, the charge causes avalanche multiplication, and an avalanche current is generated.

[0044] The quenching element **211** has a function of replacing a change in avalanche current generated in the APD **201** with a voltage signal. Further, the quenching element **211** functions as a load circuit (quenching circuit) during signal multiplication by avalanche multiplication. At this time, the quenching element **211** suppresses the voltage supplied to the APD **201** to suppress avalanche multiplication (quenching operation). The quenching element **211** may be, for example, a resistive element.

[0045] The waveform shaping unit **212** shapes a potential change of the cathode of the APD **201** generated at the time of photon detection, and outputs a pulse signal. As the waveform shaping unit **212**, for example, a circuit having a waveform shaping effect such as an inverter circuit or a buffer circuit is used.

[0046] The pulse processing unit **220** receives photon detection pulses generated by the pulse generation unit **210** and counts the number of photon detection pulses. The configuration and operation of the pulse processing unit **220** will be described below with reference to FIG. **6**.

[0047] In the present embodiment, a configuration example using a single photon avalanche diode (SPAD) by the APD **201** as the photoelectric conversion element is illustrated, but the photoelectric conversion element is not limited thereto. Any photoelectric conversion element can be applied to the photoelectric conversion device **100** of the present embodiment as long as the photoelectric conversion element has a configuration capable of reading out information on the amount of incident light obtained by photoelectric conversion during an accumulation period.

[0048] Furthermore, in the present embodiment, all of the plurality of pixels **101** in the pixel region **12** are read, but the present invention is not limited thereto. For example, a part of the pixel region **12** may be read. Alternatively, only a predetermined number of bits of a digital signal acquired by each of the plurality of pixels **101** may be read. In addition, in the present embodiment, each of the plurality of pixels **101** has a pixel memory and is capable of performing reading by a global shutter method, but the present invention is not limited thereto. For example, reading may be performed by a rolling shutter method.

[0049] FIGS. **5**A, **5**B, and **5**C are diagrams for explaining an operation of the APD **201** according to the present embodiment. FIG. **5**A is a diagram illustrating the APD **201**, the quenching element **211**, and the waveform shaping unit **212** extracted from FIG. **4**. As illustrated in FIG. **5**A, a connection node between the APD **201**, the quenching element **211**, and the input terminal of the waveform shaping unit **212** is node A. In addition, as illustrated in FIG. **5**A, the output side of the waveform shaping unit **212** is node B. In the description based on FIGS. **5**A, **5**B, and **5**C, the waveform shaping unit **212** is assumed to be an inverter circuit.

[0050] FIG. **5**B is a graph illustrating a temporal change in potential of node A of FIG. **5**A. FIG. **5**C is a graph illustrating a temporal change in potential of node B of FIG. **5**A. During a period from time t**0** to time t**1**, a voltage of VH-VL is applied to the APD **201** of FIG. **5**A. When a photon is incident on the APD **201** at time t**1**, avalanche multiplication occurs in the APD **201**. As a result, an avalanche current flows through the quenching element **211**, and the potential of node A drops. Thereafter, the potential drop amount further increases, and the voltage applied to the APD **201** gradually decreases. Then, at time t**2**, the avalanche multiplication in the APD **201** stops. As a result, the voltage level of node A does not drop below a certain fixed value. Thereafter, during a period from time t**2** to time t**3**, a current compensating for the voltage drop flows from the node of the voltage VH into node A, and node A settles to its original potential at time t**3**.

[0051] In the above-described process, the potential of node B goes to a high level during a period in which the potential of node A is lower than a certain threshold. In this way, the waveform of the potential drop of node A caused by the incidence of photon is shaped by the waveform shaping unit **212** and output as a pulse to node B.

[0052] FIG. **6** is a block diagram illustrating a configuration example of the pulse processing unit **220** according to the present embodiment. The pulse processing unit **220** includes a pixel counter **221** and a pixel memory **222** (memory). FIG. **6** illustrates an example in which the pixel counter **221** can output an 8-bit digital signal and the pixel memory **222** has an 8-bit storage capacity, but the number of bits is not limited to 8.

[0053] The pixel counter **221** counts the number of photon detection pulses output from the pulse generation unit **210**, starting from the accumulation start time of one frame period. A memory transfer signal is input to the pixel memory **222** via a transfer signal line **118**. The pixel memory **222** latches a count value held by the pixel counter **221** at the timing when the memory transfer signal is asserted. The memory transfer signal may be output from the vertical control pulse generation unit **110**, may be output from the horizontal control pulse generation unit **111**, or may be output from a control circuit that is not illustrated. The controls of the pixel counter **221** and the pixel memory **222**, such as starting an operation, stopping an operation, and resetting a held value, are performed for each row by a control signal supplied from the vertical control pulse generation unit **110** via the vertical control line **112**.

[0054] The values held in the pixel memory **222** are output to the data output line **114** during accumulation in one frame period and at a predetermined timing after the accumulation is completed. The data output line **114** is an 8-bit parallel bus extending in the first direction, and can transmit an 8-bit digital signal. A plurality of pixel memories **222** arranged in the horizontal direction is connected to the data output line **114**.

[0055] The vertical control pulse generation unit **110** and the horizontal control pulse generation unit **111** output, to the pixel memory **222**, a selection signal for selecting the pixel signal processing unit **103** that output a count value via the vertical control line **112** and the horizontal control line **113**, respectively. In response to this selection signal, electrical connection or disconnection between the pixel memory **222** and the data output line **114** is switched. As a result, the pixel memory **222** of the selected pixel signal processing unit **103** outputs the held value to the data output line **114**.

[0056] FIG. **7** is a timing chart for explaining an operation of the pulse processing unit **220** according to the present embodiment. FIG. **7** mainly illustrates an operation in frame period F**1** among three consecutive frame periods F**0**, F**1**, and F**2**.

[0057] In the present embodiment, one frame period F**1** includes four types of accumulation periods F**1_1**, F**1_2**, F**1_3**, and F**1_4**. Accumulation period F**1_1** is a period from time T**0**, which is the start time of frame period F**1**, to time T**1**. If the length of frame period F**1** is T, the length of accumulation period F**1_1** is T/4. Accumulation period F**1_2** is a period from time T**0** to time T**2**. The length of accumulation period F**1_2** is 2T/4. Accumulation period F**1_3** is a period from time T**0** to time T**3**. The length of accumulation period F**1_3** is 3T/4. Accumulation period F**1_4** is a period from time T**0** to time T**4**, which is the end time of frame period F**1**. The length of accumulation period F**1_4** is T. In other words, accumulation period F**1_4** is the same period as frame period F**1**. In this manner, accumulation period F**1_4** (second accumulation period) includes accumulation period F**1_1** (first accumulation period) and a period after accumulation period F**1_1** (time T**1** to time T**4**).

[0058] The "count value" in FIG. **7** indicates a count value held in the pixel counter **221**. The count value increases over time within frame period F**1** as the pixel counter **221** counts the number of photon detection pulses. Then, the count value is reset when the time has elapsed to each of times T**0** and T**4** at which the frame period is switched. However, the count value is not reset at a timing in the middle of frame period F**1**. That is, the count value is not reset at the end of any of accumulation periods F**1_1**, F**1_2**, and F**1_3** (time T**1**, T**2**, or T**3**). The count values at times T**1**, T**2**, T**3**, and T**4** are denoted by C**1**, C**2**, C**3**, and C**4**, respectively. In this period, the count value monotonically increases. Therefore, the count value C**2** is greater than the count value C**1**, the count value C**3** is greater than the count value C**2**, and the count value C**4** is greater than the count value C**3**.

[0059] In addition, the count values C**1**, C**2**, C**3**, and C**4** are temporarily stored in the pixel memory **222**. The "pixel memory" in FIG. **7** indicates a count value temporarily held in the pixel memory **222**. The signals temporarily stored in the pixel memory **222** is sequentially output to the signal processing unit **117** via the data output line **114** and the signal reading circuit **115**. The "output" in FIG. **7** indicates an accumulation period of a signal output from the pixel memory **222**.

[0060] As described above, a signal accumulated in accumulation period F**1_1** is read in a period from time T**1** to time T**2**. The signal processing unit **117** can start the processing at this point. Similarly, a signal accumulated in accumulation period F**1_2** is read in a period from time T**2** to time T**3**, and a signal accumulated in accumulation period F**1_3** is read in a period from time T**3** to time T**4**. A signal accumulated in accumulation period F**1_4** is read in a period after time T**4**. The signal processing unit **117** can acquire signals for one frame period F**1** a plurality of times, and can sequentially perform processing. Note that the signal processing unit **117** may not use all of the signal accumulated in accumulation period F**1_1**, the signal accumulated in accumulation period F**1_2**, the signal accumulated in accumulation period F**1_3**, and the signal accumulated in accumulation period F**1_4**. As will be described below, in the present embodiment, the signal accumulated in accumulation period F**1_1** and the signal accumulated in accumulation period F**1_4** are used for processing.

[0061] FIG. **8** is a diagram illustrating an example of an image acquired by the photoelectric conversion device according to the present embodiment. FIG. **8** illustrates images based on the signals accumulated in accumulation periods F**1_1**, F**1_2**, F**1_3**, and F**1_4** described above. The shading in FIG. **8** indicates a pixel value, and a portion closer to white has a larger pixel value. As illustrated in FIG. **8**, as the accumulation period is longer, an image with a subject (vehicle and tree) having a larger pixel value is obtained.

[0062] FIG. **9** is a block diagram illustrating a configuration example of the signal processing unit **117** according to the present embodiment. The signal processing unit **117** (information processing device) includes buffers **171** and **172**, an abnormal pixel detection unit **173**, an abnormal pixel information holding unit **174**, an abnormal pixel correction unit **175**, and a corrected image holding unit **176**.

[0063] The buffer **171** holds an image based on the signal accumulated in accumulation period F**1_1** among the plurality of images output from the pixel signal processing unit **103** of the first circuit region **22** via the signal reading circuit **115**. The buffer **172** holds an image based on the signal accumulated in accumulation period F**1_4** among the plurality of images output from the pixel signal processing unit **103** of the first circuit region **22** via the signal reading circuit **115**. That is, the buffers **171** and **172** are frame buffers each having a storage capacity capable of holding one image.

[0064] The abnormal pixel detection unit **173** detects an abnormal pixel from the image based on the signal accumulated in accumulation period F**1_1** and held in the buffer **171**. The detection of the abnormal pixel may be processing of detecting that the pixel does not satisfy a predetermined criterion. Here, the predetermined criterion is determined in consideration of a range in which the pixel is handled as having no problem, and is appropriately set in consideration of, for example, a manufacturing error of the photoelectric conversion device. Then, the abnormal pixel detection unit **173** outputs abnormal pixel information indicating an abnormal pixel detection result to the abnormal pixel information holding unit **174**. The abnormal pixel information may include pixel information indicating that the pixel does not satisfy the predetermined criterion. The abnormal pixel detection unit **173** is an example of a detection unit in the signal processing unit **117**.

[0065] The detection of the abnormal pixel in the present embodiment will be described. The abnormal pixel is a pixel that outputs a unique pixel value as compared with neighboring pixels, not a pixel value corresponding to incident light. The abnormal pixel may include a so-called white or black scratch. The white scratch is an abnormal pixel having a significantly larger pixel value than neighboring pixels, and the black scratch is an abnormal pixel having a significantly smaller pixel value than neighboring pixels. In this manner, the pixel value of the abnormal pixel is often significantly different from the pixel values of the surrounding pixels. Therefore, by comparing the pixel value of the target pixel with the pixel values of its surroundings (e.g., eight neighboring pixels), it is possible to determine whether the target pixel is an abnormal pixel based on whether the pixel value of the target pixel is significantly different.

[0066] FIG. **10** is a schematic diagram illustrating an example of a target pixel and peripheral pixels according to the present embodiment. FIG. **10** illustrates pixels of 3 rows and 3 columns extracted from the plurality of pixels **101** together with their pixel values. FIG. **10** illustrates a target pixel **101***a* arranged at the center and eight peripheral pixels **101***b* arranged around the target pixel **101***a*. A numerical value described in a circle representing each pixel indicates a pixel value. As illustrated in FIG. **10**, the pixel value of the target pixel **101***a* is significantly larger than the pixel values of the peripheral pixels **101***b*. Therefore, the abnormal pixel can be detected, for example, by calculating a difference between the pixel value of the target pixel **101***a* and the average value of the pixel values of the peripheral pixels **101***b*, and determining that the target pixel **101***a* is an abnormal pixel when the difference is a predetermined threshold or more.

[0067] Note that, as illustrated in FIG. **8**, in the present embodiment, the image in accumulation period F**1_1** used for detection of abnormal pixel is an image having a pixel value smaller than that of the image in accumulation period F**1_4** for image generation. Therefore, a value smaller than the difference between the pixel value of the target pixel **101***a* and the average value of the pixel values of the peripheral pixels **101***b* assumed in the image in accumulation period F**1_4** for image generation may be used as the threshold for determination in the abnormal pixel detection unit **173**. The length of accumulation period F**1_1** is ¼ of the length of accumulation period F**1_4**. Therefore, it is preferable that the threshold is set to a relatively small value of about ¼ of the difference between the pixel value of the target pixel **101***a* and the average value of the pixel values of the peripheral pixels **101***b* assumed in the image in accumulation period F**1_4** for image generation.

[0068] The abnormal pixel information output from the abnormal pixel detection unit **173** is held in the abnormal pixel information holding unit **174**. The abnormal pixel information may include information indicating a coordinate of an abnormal pixel among the plurality of pixels **101**. The abnormal pixel information may be map data in a matrix format having the number of rows and the number of columns of the pixels **101**. Each element of this matrix is a 1-bit value indicating whether the pixel is normal or abnormal. The abnormal pixel information may be table data

indicating XY coordinate information (row number and column number) of the abnormal pixel.
[0069] The abnormal pixel correction unit **175** corrects the abnormal pixel in the image in accumulation period F**1_4** held in the buffer **172** based on the abnormal pixel information held in the abnormal pixel information holding unit **174**. The method for correcting the abnormal pixel may be, for example, specifying a coordinate of the abnormal pixel from the abnormal pixel information, and replacing a value of the abnormal pixel at that coordinate with an interpolation value calculated from values of a plurality of peripheral pixels. The image in which the abnormal pixel is corrected by the abnormal pixel correction unit **175** is output to the corrected image holding unit **176**. The corrected image holding unit **176** temporarily holds the corrected pixel. The corrected pixel can be output from the corrected image holding unit **176** to a circuit at the subsequent stage. The abnormal pixel correction unit **175** is an example of a correction unit in the signal processing unit **117**.
[0070] FIG. **11** is a timing chart for explaining an operation of the signal processing unit **117** according to the present embodiment. FIG. **11** illustrates timings of signal input to the signal processing unit **117**, signal processing in the signal processing unit **117**, and the like. The notations of the frame period, the "pixel memory", and the "output" are substantially the same as those in FIG. **7**, and thus the description thereof is omitted.
[0071] The "detection processing" in FIG. **11** indicates a timing at which the abnormal pixel detection unit **173** performs abnormal pixel detection processing. The "information" in FIG. **11** indicates information held in the abnormal pixel information holding unit **174** and a holding timing. The "correction processing" in FIG. **11** indicates a timing at which the abnormal pixel correction unit **175** performs abnormal pixel correction processing. Note that, in the configuration of the signal processing unit **117** in FIG. **9**, buffers for holding images in accumulation periods F**1_2** and F**1_3** are not arranged, and processing using these images is not performed. Therefore, in FIG. **11**, the "pixel memory" and "output" fields do not include operation timings related to accumulation periods F**1_2** and F**1_3** and count values C**2** and C**3**.
[0072] After the output of the image (first signal) in accumulation period F**1_1** is started at time T**1**, the abnormal pixel detection unit **173** starts abnormal pixel detection processing ("detection 1" in FIG. **11**). After a predetermined time has elapsed from the start of the abnormal pixel detection processing, the abnormal pixel information holding unit **174** updates the abnormal pixel information ("abnormal pixel information 1" in FIG. **11**). After the output of the image (second signal) in accumulation period F**1_4** is started at time T**4**, the abnormal pixel correction unit **175** starts abnormal pixel correction processing ("correction 1" in FIG. **11**).
[0073] In this manner, since the period during which the abnormal pixel detection processing is performed overlaps accumulation period F**1_4**, the abnormal pixel detection processing and the accumulation are performed in parallel. Therefore, the processing time of the abnormal pixel detection processing does not affect the length of frame period F**1**, and the abnormal pixel detection processing can be performed without affecting the entire processing time. Then, since the abnormal pixel detection processing can be completed early, the abnormal pixel correction processing can be started immediately after the end of accumulation period F**1_4**. Therefore, according to the present embodiment, there are provided a photoelectric conversion device, an information processing device, a method for controlling a photoelectric conversion device, and an information processing method in which a processing delay caused by detecting and correcting a pixel that is outside a predetermined reference range is further reduced.
[0074] In addition, since accumulation period F**1_1** for abnormal pixel detection processing is shorter than accumulation period F**1_4** for image generation, signal saturation hardly occurs. Therefore, the possibility of detection omission of an abnormal pixel caused by signal saturation can be reduced.
[0075] Furthermore, in the present embodiment, the upper limit of time that can be allocated to the abnormal pixel detection processing is 3T/4, which is relatively long. In FIG. **11**, the abnormal

pixel detection processing is performed for a time of about T/4, but it is also possible to execute additional processing, which results in the total processing time longer than T/4, thereby improving the abnormal pixel detection accuracy.

[0076] An example of the additional processing will be described. When the temperature of the sensor substrate **11** is high, the number of abnormal pixels may increase, and the abnormal pixels may be adjacent to each other. FIG. **12** is a schematic diagram illustrating an example of a target pixel and peripheral pixels according to a modification of the present embodiment. FIG. **12** illustrates an example of a pixel value in a case where the temperature of the sensor substrate **11** is high, and a peripheral pixel **101**c, which is one of the eight peripheral pixels arranged around the target pixel **101**a, is an abnormal pixel. In such a case, the difference between the pixel value of the target pixel **101**a and the average value of the pixel values of the peripheral pixels **101**b and **101**c decreases, which may decrease the abnormal pixel determination accuracy.

[0077] In such a case, the abnormal pixel determination accuracy can be improved by adding processing of determining whether the abnormal pixel is an isolated point. This processing will be described. In a case where pixel values of adjacent pixels are close to each other (the difference between the pixel values is within a predetermined range), it is determined that these adjacent pixels are connected. Then, the pixels connected to each other are treated as one pixel group, and the number of pixels (connection number) for each pixel group is counted. In the example of FIG. **12**, the number of pixels having pixel values of 20 to 40 is 7 (the connection number is 7), whereas the number of pixels having pixel values of 80 to 100 is 2 (the connection number is 2). A pixel group in which the connection number is N (N is a positive integer) or less among the pixel groups after the connection processing is set as an isolated point. For example, when N=2, it is determined that the above-described two pixels having pixel values of 80 to 100 are isolated points and both are abnormal pixels. When the temperature of the sensor substrate **11** is higher and the density of abnormal pixels is higher, the threshold N for the connection number may need to be set to 3 or more. In this manner, in a case where the threshold N for the connection number is large, the range of pixels of which pixel values are to be referred to at the time of processing becomes broad, and accordingly, the processing load, such as memory access and the amount of calculation, increases. However, even in a case where the processing load increases and a long time is required for the processing, in the present embodiment, the abnormal pixel detection processing can be started early and completed early, thereby reducing the processing delay.

Second Embodiment

[0078] In the first embodiment, the signal processing unit **117** acquires one image during accumulation in frame period F**1** and performs abnormal pixel detection processing. On the other hand, in the present embodiment, a method in which the signal processing unit **117** acquires a plurality of images during accumulation in frame period F**1** and performs abnormal pixel detection processing a plurality of times will be described. In the present embodiment, the description of elements common to the first embodiment may be omitted or simplified.

[0079] FIG. **13** is a block diagram illustrating a configuration example of the signal processing unit **117** according to the present embodiment. The signal processing unit **117** includes four buffers **171**a, **171**b, **171**c, and **172**. That is, the buffer **171** in FIG. **9** is replaced with three buffers **171**a, **171**b, and **171**c in the present embodiment. The other configurations of FIG. **13** are similar to those of FIG. **9**.

[0080] The buffer **171**a holds an image based on the signal accumulated in accumulation period F**1_1** among the plurality of images output from the pixel signal processing unit **103** of the first circuit region **22** via the signal reading circuit **115**. Similarly, the buffers **171**b, **171**c, and **172** hold images based on the signals accumulated in accumulation periods F**1_2**, F**1_3**, and F**1_4**, respectively.

[0081] The abnormal pixel detection unit **173** acquires a plurality of images based on the signals accumulated in accumulation periods F**1_1**, F**1_2**, and F**1_3** from the buffers **171**a, **171**b, and **171**c.

The abnormal pixel detection unit **173** detects an abnormal pixel from each of these images. The lengths of accumulation periods F**1_1**, F**1_2**, and F**1_3** are ¼, 2/4, and ¾ of the length of accumulation period F**1_4**, respectively. Therefore, the threshold for determination used for abnormal pixel detection may be different for each of these images. These thresholds may be set to, for example, ¼, 2/4, and ¾ of the difference between the pixel value of the target pixel **101***a* and the average value of the pixel values of the peripheral pixels **101***b* assumed in the image in accumulation period F**1_4** for image generation, respectively.

[0082] The abnormal pixel detection unit **173** integrates abnormal pixel detection results from the plurality of images, respectively, to generate abnormal pixel information. The abnormal pixel information output from the abnormal pixel detection unit **173** is held in the abnormal pixel information holding unit **174**. The abnormal pixel information may be map data in a matrix format having the number of rows and the number of columns of the pixels **101**. Each element of this matrix is a 1-bit value indicating whether the pixel is normal or abnormal. The abnormal pixel information may be table data indicating XY coordinate information (row number and column number) of the abnormal pixel.

[0083] An example of a method of integrating abnormal pixel detection results from a plurality of images, respectively, will be described. As a first example, there is a method of determining an abnormal pixel based on a logical product of detection results. In this method, when an abnormal pixel is detected at the same coordinate in all of the plurality of images, the pixel at that coordinate is determined to be an abnormal pixel.

[0084] In a case where the abnormal pixel information is map data having a 1-bit value, a logical product can be acquired as follows. First, when an abnormal pixel is detected in the image in accumulation period F**1_1**, “1”, which is a value indicating abnormality, is written in the corresponding coordinate of the map data held in the abnormal pixel information holding unit **174**. In addition, it is assumed that “0”, which is a value indicating normality, is written in the other coordinates. When an abnormal pixel is detected in the images in accumulation periods F**1_2** and F**1_3**, “1” is overwritten if “1” is written in the corresponding coordinate of the map data, and “0” is written otherwise. By writing values in the map data in this order, the map data in which “1” is written only in the coordinate where the detection result indicating abnormality is obtained three times is generated. In this manner, by acquiring a logical product of the plurality of detection results and determining an abnormal pixel, the possibility of erroneous detection of an abnormal pixel can be reduced.

[0085] As a second example of a method of integrating abnormal pixel detection results, there is a method of determining an abnormal pixel based on a logical sum of the detection results. In this method, when an abnormal pixel is detected in at least one of the plurality of images, the pixel at that coordinate is determined to be an abnormal pixel. In a case where the abnormal pixel information is map data, when an abnormal pixel is detected in the images in accumulation periods F**1_1**, F**1_2**, and F**1_3**, “1” is written in that coordinate regardless of the original value. By writing values in the map data in this order, the map data in which “1” is written in the coordinate where the detection result indicating abnormality is obtained at least once among the three times is generated. In this manner, by acquiring a logical sum of the plurality of detection results and determining an abnormal pixel, the possibility of detection omission of an abnormal pixel can be reduced. Note that the detection processing may not be performed for coordinate at which an abnormal pixel has been detected once. In this case, the calculation load and the processing time can be reduced.

[0086] As a third example of a method of integrating abnormal pixel detection results, there is a method of determining an abnormal pixel based on a majority logic of the detection results. In this method, when an abnormal pixel is detected in at least two (that is, the majority) of the plurality of images, the pixel at that coordinate is determined to be an abnormal pixel. As a result, it is possible to appropriately achieve a good balance between the possibility of erroneous detection of an

abnormal pixel and the possibility of detection omission of an abnormal pixel.
[0087] FIG. **14** is a timing chart for explaining an operation of the signal processing unit **117** according to the present embodiment. FIG. **14** illustrates timings of signal input to the signal processing unit **117**, signal processing in the signal processing unit **117**, and the like. The method of notating each item in FIG. **14** is substantially the same as that in FIG. **11**, and thus the description thereof is omitted.
[0088] After the output of the image in accumulation period F**1_1** is started at time T**1**, the abnormal pixel detection unit **173** starts abnormal pixel detection processing ("detection 1" in FIG. **14**). After a predetermined time has elapsed from the start of the abnormal pixel detection processing, the abnormal pixel information holding unit **174** updates the abnormal pixel information ("information 1" in FIG. **14**). After the output of the image in accumulation period F**1_2** is started at time T**2**, the abnormal pixel detection unit **173** starts abnormal pixel detection processing ("detection 2" in FIG. **14**). After a predetermined time has elapsed from the start of the abnormal pixel detection processing, the abnormal pixel information holding unit **174** updates the abnormal pixel information ("information 2" in FIG. **14**). After the output of the image in accumulation period F**1_3** is started at time T**3**, the abnormal pixel detection unit **173** starts abnormal pixel detection processing ("detection 3" in FIG. **14**). After a predetermined time has elapsed from the start of the abnormal pixel detection processing, the abnormal pixel information holding unit **174** updates the abnormal pixel information ("information 3" in FIG. **14**). After the output of the image in accumulation period F**1_4** is started at time T**4**, the abnormal pixel correction unit **175** starts abnormal pixel correction processing ("correction" in FIG. **14**).
[0089] In this manner, the plurality of times of abnormal pixel detection processing is performed at least partially in parallel with the accumulation operation in accumulation period F**1_4**. Therefore, even in a case where the abnormal pixel detection processing is performed a plurality of times, the processing time does not significantly affect frame period F**1**, and the abnormal pixel detection processing can be performed without significantly affecting the entire processing time. In addition, by integrating a plurality of detection results, the abnormal pixel detection accuracy can be improved.
[0090] Therefore, according to the present embodiment, there are provided a photoelectric conversion device, an information processing device, a method for controlling a photoelectric conversion device, and an information processing method capable of obtaining effects similar to those of the first embodiment. Furthermore, according to the present embodiment, by integrating a plurality of detection results, the abnormal pixel detection accuracy can be improved.
Third Embodiment
[0091] In the first embodiment and the second embodiment, the signal processing unit **117** has a configuration including buffers that hold images during accumulation and an abnormal pixel information holding unit **174** that holds abnormal pixel information. On the other hand, in the present embodiment, a configuration example in which an abnormal pixel is detected and corrected in real time without using these storage units will be described. In the present embodiment, the description of elements common to the first embodiment or the second embodiment may be omitted or simplified.
[0092] FIG. **15** is a block diagram illustrating a configuration example of the signal processing unit **117** according to the present embodiment. In FIG. **15**, the buffers **171** and **172** and the abnormal pixel information holding unit **174** are not arranged as compared with the configuration of the signal processing unit **117** in FIG. **9**. That is, in the present embodiment, a frame buffer having a storage capacity capable of holding one frame image is not arranged. Data input to the signal processing unit **117** via the signal reading circuit **115** is input to the abnormal pixel detection unit **173** and the abnormal pixel correction unit **175** via a plurality of data paths **177** and **178** in raster scan order. The abnormal pixel detection unit **173** and the abnormal pixel correction unit **175** include a hardware calculator that processes an input image in a predetermined clock cycle and a

delay buffer that holds data for several lines of one image. The hardware calculator and the delay buffer sequentially process and output images while absorbing a difference in processing time. In this manner, the signal processing unit **117** is configured to generate abnormal pixel information by sequentially processing a plurality of signals constituting one frame image, rather than temporarily storing the plurality of signals simultaneously, at the time of acquiring the signals from the signal reading circuit **115**.

[0093] Data based on the signal accumulated in accumulation period F**1_1** among the plurality of images output from the pixel signal processing unit **103** of the first circuit region **22** via the signal reading circuit **115** passes through the data path **177**, and is input to the abnormal pixel detection unit **173**. Data based on the signal accumulated in accumulation period F**1_4** among the plurality of images output from the pixel signal processing unit **103** of the first circuit region **22** via the signal reading circuit **115** passes through the data path **178**, and is input to the abnormal pixel correction unit **175**. The abnormal pixel detection unit **173** detects an abnormal pixel based on the input image in accumulation period F**1_1**, and outputs abnormal pixel information to the abnormal pixel correction unit **175**. Based on the abnormal pixel information, the abnormal pixel correction unit **175** corrects the abnormal pixel included in the input image in accumulation period F**1_4** in real time, and outputs the corrected image to the corrected image holding unit **176**.

[0094] FIG. **16** is a timing chart for explaining an operation of the signal processing unit **117** according to the present embodiment. FIG. **16** illustrates timings of signal input to the signal processing unit **117**, signal processing in the signal processing unit **117**, and the like. The method of notating each item in FIG. **16** is substantially the same as that in FIG. **11**, and thus the description thereof is omitted.

[0095] At time T**1**, an image in accumulation period F**1_1** is held in the pixel memory **222**. In the present embodiment, since no frame buffer capable of holding one frame image is arranged in the signal processing unit **117**, the image in accumulation period F**1_1** is not output and remains held in the pixel memory **222** between time T**1** and time T**3**. At time T**3**, after the output of the image in accumulation period F**1_1** is started, the abnormal pixel detection unit **173** starts abnormal pixel detection processing (“detection 1” in FIG. **16**). After a predetermined time has elapsed from the start of the abnormal pixel detection processing, the abnormal pixel detection unit **173** updates the abnormal pixel information to be output to the abnormal pixel correction unit **175** (“information 1” in FIG. **16**). Abnormal pixel information 1 is updated after the predetermined time elapses. After the output of the image in accumulation period F**1_4** is started at time T**4**, the abnormal pixel correction unit **175** starts abnormal pixel correction processing (“correction” in FIG. **16**) using the abnormal pixel information output from the abnormal pixel detection unit **173**.

[0096] According to the present embodiment, there are provided a photoelectric conversion device, an information processing device, a method for controlling a photoelectric conversion device, and an information processing method capable of obtaining effects similar to those of the first embodiment. Furthermore, according to the present embodiment, since a frame buffer capable of holding a frame image is not arranged and abnormal pixel detection processing and abnormal pixel correction processing are performed in real time, the storage capacity of the memory used in the signal processing unit **117** can be reduced.

Fourth Embodiment

[0097] In the first to third embodiments, an abnormal pixel is detected and the abnormal pixel is corrected every time an image is input to the signal processing unit **117**. On the other hand, in the present embodiment, a configuration example in which an abnormal pixel is detected at a timing when a predetermined situation change occurs and abnormal pixel information is updated will be described. In the present embodiment, the description of elements common to any of the first to third embodiments may be omitted or simplified.

[0098] FIG. **17** is a block diagram illustrating a configuration example of the signal processing unit **117** according to the present embodiment. The signal processing unit **117** of FIG. **17** further includes a temperature detection unit **181**, a time detection unit **182**, and a control unit **183** in addition to the configuration of the signal processing unit **117** of FIG. **9**. In addition, the signal processing unit **117** in FIG. **17** includes a first information holding unit **184** and a second information holding unit **185**. The first information holding unit **184** and the second information holding unit **185** correspond to the abnormal pixel information holding unit **174** in FIG. **9**.

[0099] In the present embodiment, abnormal pixel information is held by both a nonvolatile memory and a volatile memory. The first information holding unit **184** includes a nonvolatile memory such as a flash memory. The first information holding unit **184** stores in advance abnormal pixel information detected during inspection or the like at the time of factory shipment or abnormal pixel information acquired during past operations. The second information holding unit **185** includes a volatile memory such as SRAM or DRAM. The second information holding unit **185** temporarily stores abnormal pixel information acquired by abnormal pixel detection processing. The abnormal pixel information stored in the second information holding unit **185** is appropriately updated when abnormal pixel detection processing is performed. In addition, the abnormal pixel information held in the second information holding unit **185** disappears when the power supply of the photoelectric conversion device **100** is turned off.

[0100] The temperature detection unit **181** acquires temperature information from a temperature sensor arranged in the sensor substrate **11**, and outputs the temperature information to the control unit **183**. As a result, the control unit **183** can detect a change in temperature of the pixel **101**.

[0101] The time detection unit **182** acquires time information corresponding to a cumulative operating time, and outputs the time information to the control unit **183**. This time information may be acquired, for example, based on the number of times of avalanche multiplication detected by a pulse detection sensor arranged in the first circuit region **22**, or may be acquired based on an operation time of the photoelectric conversion device **100**. As a result, the control unit **183** can detect an amount of change in cumulative operating time.

[0102] The control unit **183** controls the abnormal pixel detection unit **173**, the first information holding unit **184**, and the second information holding unit **185** based on the temperature information acquired by the temperature detection unit **181** and the time information acquired by the time detection unit **182**.

[0103] The abnormal pixel correction unit **175** acquires the abnormal pixel information stored in the second information holding unit **185**. Based on the abnormal pixel information, the abnormal pixel correction unit **175** corrects an abnormal pixel in an image output from the buffer **172**, and outputs the corrected image to the corrected image holding unit **176**.

[0104] FIG. **18** is a flowchart for explaining an operation of the photoelectric conversion device **100** according to the present embodiment. FIG. **18** illustrates a processing procedure from the start of the photoelectric conversion device **100** to the end of the operation.

[0105] In step S**01** immediately after the photoelectric conversion device **100** is started, the control unit **183** performs control to load abnormal pixel information from the first information holding unit **184** including a nonvolatile memory to the second information holding unit **185** including a volatile memory. As a result, the abnormal pixel correction unit **175** can acquire the abnormal pixel information.

[0106] In step S**02**, the control unit **183** acquires time information from the time detection unit **182**, and determines whether the amount of change in cumulative operating time of the first circuit region **22** exceeds a predetermined threshold. When the amount of change in cumulative operating time exceeds the threshold (YES in step S**02**), the process proceeds to step S**04**. When the amount of change in cumulative operating time does not exceed the threshold (NO in step S**02**), the process proceeds to step S**03**.

[0107] In step S**04**, the control unit **183** outputs a control signal for instructing the abnormal pixel detection unit **173** to perform abnormal pixel detection processing. The abnormal pixel detection unit **173** performs abnormal pixel detection processing similar to that described in the first

embodiment, and outputs abnormal pixel information to the second information holding unit **185**.
[0108] In step S**05**, the second information holding unit **185** updates the abnormal pixel information held in the volatile memory. Thereafter, in step S**06**, the first information holding unit **184** acquires the abnormal pixel information held in the volatile memory of the second information holding unit **185**, and updates the abnormal pixel information held in the nonvolatile memory. Thereafter, the process proceeds to step S**09**.
[0109] In step S**03**, the control unit **183** acquires temperature information from the temperature detection unit **181**, and determines whether the amount of change in temperature exceeds a predetermined threshold. When the amount of change in temperature exceeds the threshold (YES in step S**03**), the process proceeds to step S**07**. When the amount of change in temperature does not exceed the threshold (NO in step S**03**), the process proceeds to step S**09**.
[0110] In step S**07**, the control unit **183** outputs a control signal for instructing the abnormal pixel detection unit **173** to perform abnormal pixel detection processing. The abnormal pixel detection unit **173** performs abnormal pixel detection processing similar to that described in the first embodiment, and outputs abnormal pixel information to the second information holding unit **185**.
[0111] In step S**08**, the second information holding unit **185** updates the abnormal pixel information held in the volatile memory. Thereafter, the process proceeds to step S**09**.
[0112] The operations of subsequent steps S**09** to S**12** are generally similar to the abnormal pixel correction processing described in the first embodiment. In step S**09**, the abnormal pixel correction unit **175** acquires the abnormal pixel information held in the volatile memory of the second information holding unit **185**. In step S**10**, the abnormal pixel correction unit **175** acquires an image from the buffer **172**. In step S**11**, the abnormal pixel correction unit **175** corrects an abnormal pixel of the acquired image based on the abnormal pixel information, and outputs the corrected image to the corrected image holding unit **176**. In step S**12**, the corrected image holding unit **176** outputs the corrected image to a circuit at the subsequent stage.
[0113] In step S**13**, the control unit **183** determines whether it is time to terminate the imaging operation of the photoelectric conversion device **100**. When it is time to terminate the imaging operation (YES in step S**13**), the process of FIG. **18** ends. When it is not time to terminate the imaging operation (NO in step S**13**), the process proceeds to step S**02**, and the same abnormal pixel detection processing and abnormal pixel correction processing are repeated.
[0114] In this manner, in the present embodiment, an abnormal pixel is detected and updated according to the amount of change in temperature or the amount of change in cumulative operating time. An effect obtained by performing such an operation will be described.
[0115] In general, the number of abnormal pixels depends on the temperature of the sensor substrate **11** on which the pixels **101** are arranged. For example, when the temperature of the sensor substrate **11** rises, the number of abnormal pixels tends to increase as compared with that when the temperature is low. Therefore, if abnormal pixels are corrected at a high temperature using abnormal pixel information detected in a low temperature state, correction omission may occur due to an increase in the number of abnormal pixels accompanying the rise in temperature. Therefore, in the present embodiment, when a change in temperature exceeding the predetermined threshold is detected by the temperature detection unit **181**, the abnormal pixel detection unit **173** detects an abnormal pixel and updates the abnormal pixel information in the second information holding unit **185**.
[0116] In addition, when the cumulative operating time of the photoelectric conversion device **100** increases, the number of abnormal pixels may increase due to aging deterioration or the like. Therefore, if abnormal pixels of the photoelectric conversion device **100** are corrected using old abnormal pixel information, correction omission may occur due to an increase in the number of abnormal pixels due to aging deterioration or the like. Therefore, in the present embodiment, when a lapse of a predetermined cumulative operating time is detected by the time detection unit **182**, the abnormal pixel detection unit **173** detects an abnormal pixel and updates the abnormal pixel

information in the first information holding unit **184** and the second information holding unit **185**.
[0117] As a result, the abnormal pixel information is updated when it is effective to update the abnormal pixel information due to a change in temperature or a lapse of time, and otherwise, the abnormal pixel detection processing is omitted, thereby reducing the system load and power consumption required for abnormal pixel detection.
[0118] Note that, since the temperature of the sensor substrate **11** changes in a relatively short cycle depending on the operating time of the photoelectric conversion device **100** and the surrounding environment in which the photoelectric conversion device **100** is used, the abnormal pixel information may also be updated in a relatively short cycle. Therefore, it is preferable that the update of the abnormal pixel information accompanying the change in temperature is applied only to the volatile memory of the second information holding unit **185**. On the other hand, a change of an abnormal pixel caused by a change over time or the like occurs in a relatively long cycle and is irreversible. Therefore, it is preferable that the update of the abnormal pixel information with the lapse of time is also applied to the nonvolatile memory of the first information holding unit **184**. As a result, the same abnormal pixel information can be reused even after the photoelectric conversion device **100** is restarted.
[0119] According to the present embodiment, there are provided a photoelectric conversion device, an information processing device, a method for controlling a photoelectric conversion device, and an information processing method capable of obtaining effects similar to those of the first embodiment. Furthermore, according to the present embodiment, the system load and power consumption required for abnormal pixel detection can be reduced.

Fifth Embodiment

[0120] The photoelectric conversion device **100** of the above embodiments can be applied to various equipment. Examples of the equipment include a digital still camera, a digital camcorder, a camera head, a copying machine, a facsimile, a mobile phone, a vehicle-mounted camera, an observation satellite, and a surveillance camera. FIG. **19** is a block diagram of a digital still camera as an example of equipment. FIG. **19** illustrates an example in which the photoelectric conversion device **100** is applied to a digital still camera.
[0121] The equipment **70** illustrated in FIG. **19** includes a barrier **706**, a lens **702**, an aperture **704**, and an imaging device **700** (an example of the photoelectric conversion device **100**). The equipment **70** further includes a signal processing unit (processing device) **708**, a timing generation unit **720**, a general control/operation unit **718** (control device), a memory unit **710** (storage device), a storage medium control I/F unit **716**, a storage medium **714**, and an external I/F unit **712**. The information processing device **30** of the above-described embodiments may be included in the imaging device **700** or may be included in the signal processing unit **708**. At least one of the barrier **706**, the lens **702**, and the aperture **704** is an optical device corresponding to the equipment. The barrier **706** protects the lens **702**, and the lens **702** forms an optical image of an object on the imaging device **700**. The aperture **704** varies the amount of light passing through the lens **702**. The imaging device **700** is configured as in the above embodiments, and converts an optical image formed by the lens **702** into image data (image signal). The signal processing unit **708** performs various corrections, data compression, and the like on the image data output from the imaging device **700**. The timing generation unit **720** outputs various timing signals to the imaging device **700** and the signal processing unit **708**. The general control/operation unit **718** controls the entire digital still camera, and the memory unit **710** temporarily stores image data. The storage medium control I/F unit **716** is an interface for storing or reading image data on the storage medium **714**, and the storage medium **714** is a detachable storage medium such as a semiconductor memory for storing or reading image data. The external I/F unit **712** is an interface for communicating with an external computer or the like. The timing signal or the like may be input from the outside of the equipment. The equipment **70** may further include a display device (a monitor, an electronic view finder, or the like) for displaying information obtained by the photoelectric conversion device. The

equipment includes at least a photoelectric conversion device. Further, the equipment **70** includes at least one of an optical device, a control device, a processing device, a display device, a storage device, and a mechanical device that operates based on information obtained by the photoelectric conversion device. The mechanical device is a movable portion (for example, a robot arm) that receives a signal from the photoelectric conversion device for operation.

[0122] Each pixel may include a plurality of photoelectric conversion units (a first photoelectric conversion unit and a second photoelectric conversion unit). The signal processing unit **708** may be configured to process a pixel signal based on charges generated in the first photoelectric conversion unit and a pixel signal based on charges generated in the second photoelectric conversion unit, and acquire distance information from the imaging device **700** to an object.

Sixth Embodiment

[0123] FIGS. **20**A and **20**B are block diagrams of equipment relating to the vehicle-mounted camera according to the present embodiment. FIGS. **20**A and **20**B illustrate an example in which the photoelectric conversion device **100** is applied to a movable body such as a vehicle. The equipment **80** includes an imaging device **800** (an example of the photoelectric conversion device **100** or a photoelectric conversion system) and a signal processing device (processing device) that processes a signal from the imaging device **800**. The equipment **80** includes an image processing unit **801** that performs image processing on a plurality of pieces of image data acquired by the imaging device **800**, and a parallax calculation unit **802** that calculates parallax (phase difference of parallax images) from the plurality of pieces of image data acquired by the equipment **80**. The information processing device **30** of the above-described embodiments may be included in the imaging device **800** or may be included in the image processing unit **801**. The equipment **80** includes a distance measurement unit **803** that calculates a distance to an object based on the calculated parallax, and a collision determination unit **804** that determines whether or not there is a possibility of collision based on the calculated distance. Here, the parallax calculation unit **802** and the distance measurement unit **803** are examples of a distance information acquisition unit that acquires distance information to the object. That is, the distance information is information on a parallax, a defocus amount, a distance to the object, and the like. The collision determination unit **804** may determine the possibility of collision using any of these pieces of distance information. The distance information acquisition unit may be realized by dedicatedly designed hardware or software modules. Further, it may be realized by a field programmable gate array (FPGA), an application specific integrated circuit (ASIC) or a combination thereof.

[0124] The equipment **80** is connected to the vehicle information acquisition device **810**, and can obtain vehicle information such as a vehicle speed, a yaw rate, and a steering angle. Further, the equipment **80** is connected to a control ECU **820** which is a control device that outputs a control signal for generating a braking force to the vehicle based on the determination result of the collision determination unit **804**. The equipment **80** is also connected to an alert device **830** that issues an alert to the driver based on the determination result of the collision determination unit **804**. For example, when the collision possibility is high as the determination result of the collision determination unit **804**, the control ECU **820** performs vehicle control to avoid collision or reduce damage by braking, returning an accelerator, suppressing engine output, or the like. The alert device **830** alerts the user by sounding an alarm such as a sound, displaying alert information on a screen of a car navigation system or the like, or giving vibration to a seat belt or a steering wheel. The equipment **80** functions as a control unit that controls the operation of controlling the vehicle as described above.

[0125] In the present embodiment, an image of the periphery of the vehicle, for example, the front or the rear is captured by the equipment **80**. FIG. **20**B illustrates equipment in a case where an image is captured in front of the vehicle (image capturing range **850**). The vehicle information acquisition device **810** as the imaging control unit sends an instruction to the equipment **80** or the imaging device **800** to perform the imaging operation. With such a configuration, the accuracy of

distance measurement can be further improved.
[0126] Although the example of control for avoiding a collision to another vehicle has been described above, the embodiment is applicable to automatic driving control for following another vehicle, automatic driving control for not going out of a traffic lane, or the like. Furthermore, the equipment is not limited to a vehicle such as an automobile and can be applied to a movable body (movable apparatus) such as a ship, an airplane, a satellite, an industrial robot and a consumer use robot, or the like, for example. In addition, the equipment can be widely applied to equipment which utilizes object recognition or biometric authentication, such as an intelligent transportation system (ITS), a surveillance system, or the like without being limited to movable bodies.
Modified Embodiments
[0127] The present invention is not limited to the above embodiments, and various modifications are possible. For example, an example in which some of the configurations of any one of the embodiments are added to other embodiments or an example in which some of the configurations of any one of the embodiments are replaced with some of the configurations of other embodiments are also embodiments of the present invention.
[0128] The disclosure of this specification includes a complementary set of the concepts described in this specification. That is, for example, if a description of “A is B” (A=B) is provided in this specification, this specification is intended to disclose or suggest that “A is not B” even if a description of “A is not B” (A≠B) is omitted. This is because it is assumed that “A is not B” is considered when “A is B” is described.
[0129] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
[0130] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
[0131] This application claims the benefit of Japanese Patent Application No. 2024-029439, filed Feb. 29, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A photoelectric conversion device comprising: a pixel configured to generate a signal corresponding to an amount of incident light; a memory configured to hold the signal; and a processing unit configured to perform signal processing based on the signal held in the memory, wherein the pixel generates the signal in a second accumulation period including a first accumulation period and a period after the first accumulation period, wherein the processing unit acquires a first signal corresponding to an amount of incident light in the first accumulation period from the memory in a period from an end of the first accumulation period to an end of the second accumulation period, wherein the processing unit generates pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using the first signal, wherein the processing unit acquires a second signal corresponding to an amount of incident light in the second accumulation period from the memory after the end of the second accumulation period, and wherein the processing unit corrects the second signal based on the pixel information.

2. The photoelectric conversion device according to claim 1, wherein a period in which the processing unit detects that the pixel does not satisfy the predetermined criterion at least partially overlaps the second accumulation period.

3. The photoelectric conversion device according to claim 1, wherein the pixel includes an avalanche photodiode, and wherein the memory holds a count value based on a photon incident on the avalanche photodiode as the signal.

4. The photoelectric conversion device according to claim 3, wherein the count value is not reset from a start of the second accumulation period to the end of the second accumulation period.

5. The photoelectric conversion device according to claim 1, wherein the photoelectric conversion device includes a plurality of the pixels, and wherein the pixel information includes information indicating a coordinate of a pixel detected not to satisfy the predetermined criterion among the plurality of pixels.

6. The photoelectric conversion device according to claim 5, wherein the processing unit corrects the second signal corresponding to the pixel at the coordinate.

7. The photoelectric conversion device according to claim 1, wherein the second accumulation period includes a plurality of the first accumulation periods, wherein the processing unit acquires a plurality of the first signals during a period from a start of the second accumulation period to the end of the second accumulation period, and wherein the processing unit generates the pixel information based on the plurality of first signals.

8. The photoelectric conversion device according to claim 7, wherein the processing unit generates a detection result of a 1-bit value based on each of the plurality of first signals, and generates the pixel information based on a logical product of a plurality of the detection results.

9. The photoelectric conversion device according to claim 7, wherein the processing unit generates a detection result of a 1-bit value based on each of the plurality of first signals, and generates the pixel information based on a logical sum of a plurality of the detection results.

10. The photoelectric conversion device according to claim 7, wherein the processing unit generates a detection result of a 1-bit value based on each of the plurality of first signals, and generates the pixel information based on a majority logic of a plurality of the detection results.

11. The photoelectric conversion device according to claim 7, wherein the processing unit generates the pixel information further based on a plurality of different thresholds set to correspond to the plurality of first signals, respectively.

12. The photoelectric conversion device according to claim 1, the photoelectric conversion device includes a plurality of the pixels, and the processing unit generates the pixel information by sequentially processing a plurality of the first signals output from the plurality of pixels, respectively, rather than temporarily storing the plurality of first signals simultaneously.

13. The photoelectric conversion device according to claim 1 further comprising: a holding unit configured to hold the pixel information; and a temperature detection unit configured to detect a temperature of the pixel, wherein the holding unit updates the pixel information based on an amount of change in temperature.

14. The photoelectric conversion device according to claim 1, further comprising: a holding unit configured to hold the pixel information; and a time detection unit configured to detect a cumulative operating time of the photoelectric conversion device, wherein the holding unit updates the pixel information based on an amount of change in cumulative operating time.

15. The photoelectric conversion device according to claim 14, wherein the holding unit includes a nonvolatile memory, and wherein the holding unit updates the pixel information held in the nonvolatile memory.

16. Equipment comprising: the photoelectric conversion device according to claim 1; and at least any one of: an optical device adapted for the photoelectric conversion device, a control device configured to control the photoelectric conversion device, a processing device configured to process a signal output from the photoelectric conversion device, a display device configured to display information obtained by the photoelectric conversion device, a storage device configured to store information obtained by the photoelectric conversion device, and a mechanical device configured to operate based on information obtained by the photoelectric conversion device.

17. The equipment according to claim 16, wherein the processing device acquires distance information on a distance from the photoelectric conversion device to an object.

18. An information processing device comprising: a detection unit configured to generate pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using a first signal corresponding to an amount of light incident on the pixel in a first accumulation period; and a correction unit configured to correct a second signal corresponding to an amount of light incident on the pixel in a second accumulation period including the first accumulation period and a period after the first accumulation period based on the pixel information, wherein the detection unit acquires the first signal during a period from an end of the first accumulation period to an end of the second accumulation period, and wherein the correction unit acquires the second signal after the end of the second accumulation period.

19. A method for controlling a photoelectric conversion device including a pixel configured to generate a signal corresponding to an amount of incident light, a memory configured to hold the signal, and a processing unit configured to perform signal processing based on the signal held in the memory, the method comprising: generating, by the pixel, the signal in a second accumulation period including a first accumulation period and a period after the first accumulation period; acquiring, by the processing unit, a first signal corresponding to an amount of incident light in the first accumulation period from the memory in a period from an end of the first accumulation period to an end of the second accumulation period; generating, by the processing unit, pixel information indicating that a pixel does not satisfy a predetermined criterion when it is detected that the pixel does not satisfy the predetermined criterion by using the first signal; acquiring, by the processing unit, a second signal corresponding to an amount of incident light in the second accumulation period from the memory after the end of the second accumulation period; and correcting, by the processing unit, the second signal based on the pixel information.

20. An information processing method comprising: generating abnormal pixel information indicating an abnormal pixel detection result based on a first signal corresponding to an amount of light incident on a pixel in a first accumulation period; and correcting a second signal corresponding to an amount of light incident on the pixel in a second accumulation period including the first accumulation period and a period after the first accumulation period based on the abnormal pixel information, wherein the first signal is acquired during a period from an end of the first accumulation period to an end of the second accumulation period, and wherein the second signal is acquired after the end of the second accumulation period.